BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

EEE 316 (January 2024)

Laboratory on Power Electronics

Final Project Report

Section: B2 Group: 06

SPWM Inverter using Decade Counter 4017 IC

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Signature of Instructor: _____

Academic Honesty Statement:

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"In signing this statement, We hereby certify that the work on this project is our own and that we have not copied the work of any other students (past or present), and cited all relevant sources while completing this project. We understand that if we fail to honor this agreement, We will each receive a score of ZERO for this project and be subject to failure of this course."

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1 Abstract

This project focuses on the design and implementation of a DC-to-AC inverter utilizing Sinusoidal Pulse Width Modulation (SPWM) techniques. The system is built around the 4017 IC decade counter, with a 3524 IC used to generate the required pulse signals. A pure DC signal serves as the input, which is modulated using the 3524 IC to produce pulse-width-modulated signals. These signals are passed through a buffer circuit for smoothing and then fed to MOSFETs for inversion. The real and inverted signals are processed through the decade counter to ensure that only one counter output is active at any given time. The SPWM signals are then directed to a push-pull inverter circuit using specific counter pins (1, 3, 4, 5, and 7) to achieve efficient pulse-width modulation. This design provides a cost-effective and efficient method for generating AC signals from a DC source, making it suitable for various power electronics applications.

2 Introduction

Pulse Width Modulation (PWM) is a technique used to control the output voltage by modulating the width (or duration) of a periodic pulse applied to an Inverter switch.
This technique is used for:
1.Adjustable output Voltage and Power according to Pulse width.
2.Generation of High quality Waveform. (Closely approximate sine wave) .
3.Controlling Motor speed and torque to drive mechanical device.
4.Frequency adjustment.

Inverters play a crucial role in modern power electronics by converting DC power into AC power for use in various applications such as renewable energy systems, uninterruptible power supplies, and motor drives. The focus of this project is on the development of a Sinusoidal Pulse Width Modulation (SPWM)-based DC-to-AC inverter. SPWM is a widely used technique in inverter circuits due to its ability to reduce harmonic distortion and improve output waveform quality.

The design utilizes a 4017 IC decade counter and a 3524 IC as the core components for generating and managing the SPWM signals. The input is a pure DC signal, which is modulated through the 3524 IC to generate pulse-width-modulated signals. These signals are smoothed using a buffer circuit and then inverted using MOSFETs. The real and inverted signals are fed into the 4017 IC to control the output, ensuring that only one output is active at a time. This arrangement simplifies the circuit while maintaining high efficiency and performance.

The output SPWM signals are directed to a push-pull inverter circuit, with pins 1, 3, 4, 5, and 7 of the decade counter utilized for effective pulse-width modulation. The resulting design is a robust and efficient inverter circuit capable of delivering high-quality AC output, making it ideal for educational and practical applications in power electronics.

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

The scope of this project encompasses the design, development, and testing of a Sinusoidal Pulse Width Modulation (SPWM) DC-to-AC inverter using a 4017 IC decade counter. The project aims to provide a cost-effective and efficient solution for converting DC power into high-quality AC power, with applications in various fields, including renewable energy systems, motor drives, and uninterruptible power supplies.

This project specifically addresses the following:

1. **Signal Generation and Modulation**: Using a 3524 IC to generate pulse-width-modulated signals from a pure DC input, ensuring an efficient and precise modulation process.
2. **Circuit Efficiency**: Employing a buffer circuit for signal smoothing and MOSFETs for inversion to optimize circuit performance and reduce energy losses.
3. **Output Control**: Leveraging the 4017 IC decade counter to manage SPWM signal output, ensuring that only one counter output is active at any given time. This approach minimizes switching losses and enhances the stability of the inverter.
4. **SPWM Output Application**: Driving a push-pull inverter with carefully selected decade counter pins (1, 3, 4, 5, and 7) to generate a high-quality AC signal with reduced harmonic distortion.

The project aims to serve as a practical educational model while also exploring its potential applications in real-world scenarios. The modular design allows for easy scalability and adaptability to various power levels and configurations, making it a versatile solution for a wide range of power electronics applications.

3.1.2 Literature Review

The development of efficient DC-to-AC inverters has been a key focus in power electronics, driven by the increasing demand for renewable energy systems and portable electronic devices. Various studies and innovations in Sinusoidal Pulse Width Modulation (SPWM) techniques have demonstrated their effectiveness in reducing harmonic distortion and improving inverter performance. This literature review explores the foundational concepts and technological advancements relevant to the design and implementation of SPWM DC-to-AC inverters.

SPWM Techniques

SPWM is widely recognized as a superior modulation technique for inverters due to its ability to produce a nearly sinusoidal AC output while maintaining high efficiency. Research by Bose (1990) highlights SPWM as a pivotal method for minimizing total harmonic distortion (THD) in inverter output, making it ideal for applications requiring high-quality power conversion. Subsequent studies have refined SPWM algorithms to enhance the precision of output waveforms and improve the performance of inverters under varying load conditions.

Role of Decade Counters in Inverter Circuits

The 4017 IC decade counter is frequently utilized in power electronics for its simplicity and reliability in signal sequencing and control. Literature indicates its application in controlling switching sequences in inverter circuits, ensuring accurate timing and reducing the risk of simultaneous switching. For instance, Mehta and Mehta (2005) demonstrated the use of decade counters in push-pull inverters to streamline circuit design and enhance operational stability.

Buffer Circuits and MOSFET Switching

Buffer circuits are critical in ensuring signal integrity during modulation and inversion processes. Studies on buffer circuit design emphasize their role in reducing signal noise and stabilizing voltage levels. MOSFETs, on the other hand, are preferred in inverter circuits due to their fast switching capabilities and low conduction losses. Research by Rashid (2014) underscores the advantages of using MOSFETs in SPWM inverters, particularly in high-frequency applications.

Push-Pull Inverters

Push-pull inverter topologies are widely used for their simplicity and efficiency in power conversion. Previous work has explored their integration with SPWM techniques to achieve optimal performance. According to Mohan et al. (2003), push-pull configurations offer a balanced approach to handling power delivery while maintaining waveform quality.

Challenges and Innovations

Despite advancements, challenges such as switching losses, harmonic distortion, and thermal management persist in inverter design. Recent innovations, including the integration of digital controllers and improved semiconductor materials, have addressed these issues to some extent. However, studies suggest that optimizing the interaction between circuit components remains crucial for achieving further efficiency gains.

Conclusion

The reviewed literature provides a solid foundation for the design of an SPWM DC-to-AC inverter using a 4017 IC decade counter. The integration of SPWM techniques, buffer circuits, MOSFETs, and push-pull topology aligns with established best practices in inverter design. This

project builds upon these concepts to develop a robust and efficient inverter circuit, contributing to ongoing research and application in power electronics.

3.1.3 Formulation of Problem

The need for efficient and cost-effective DC-to-AC conversion is critical in power electronics, particularly for applications such as renewable energy systems, uninterruptible power supplies, and motor drives. Conventional inverter designs often suffer from high harmonic distortion, energy losses, and complex circuitry.

This project aims to address these challenges by designing a Sinusoidal Pulse Width Modulation (SPWM) DC-to-AC inverter using a 4017 IC decade counter. The primary objectives are to:

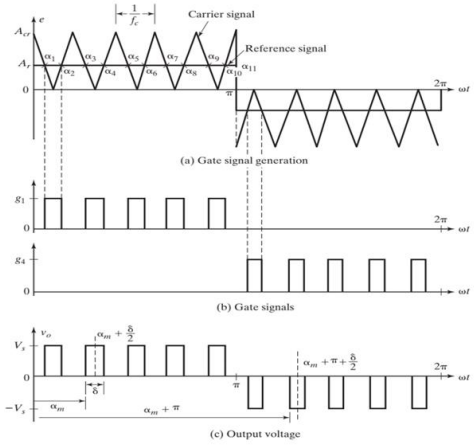
1. Minimize harmonic distortion in the output waveform.
2. Simplify circuit design while maintaining high efficiency.
3. Ensure stable and reliable operation with reduced switching losses.

The proposed solution leverages SPWM techniques, MOSFETs, and a push-pull topology to deliver high-quality AC output suitable for various power electronics applications.

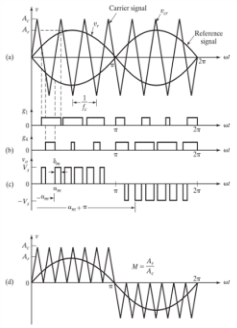
3.1.4 Analysis

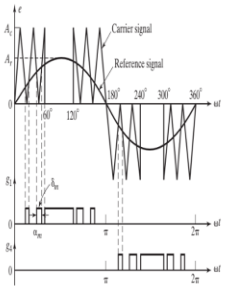
Let’s analyze first what SPWM is and how it is done traditionally.
SPWM (Sinusoidal Pulse Width Modulation) is achieved by comparing a sinusoidal reference signal with a triangular carrier wave, where the points of intersection between the two waveforms generate switching signals that control the width of pulses in an inverter, effectively creating an output voltage that closely resembles a sine wave by varying the pulse widths according to the amplitude of the reference signal at each point in time; essentially, the wider the pulse, the higher the voltage at that point in the cycle.

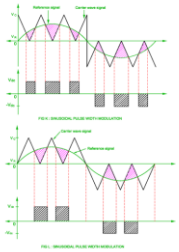
Multiple Pulse width Modulation(uniform Pulse width)

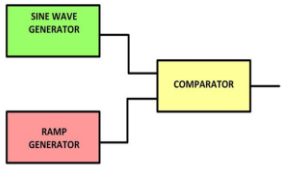


Sinusoidal-Pulse-Width Modulation









Ultimate goal is generate sine wave.
Why not a step up transformer at the end of sine wave generator used?

<u>But we did not follow this traditional process whereas we did the project following the below procedure:</u>

Input and SPWM Signal Generation

- The inverter begins with a pure DC input. This DC signal is fed into the 3524 IC, which generates pulse-width-modulated (PWM) signals.
- The SPWM technique ensures that the pulses vary in width according to a sinusoidal reference waveform, allowing for the generation of an AC-like waveform after inversion.

Signal Smoothing and Buffering

- The PWM signals from the 3524 IC are passed through a buffer circuit to smooth out any noise or irregularities.
- The buffer also amplifies the signals to drive the next stage effectively.

Signal Inversion Using MOSFETs

- The smoothed PWM signals are fed into a MOSFET driver circuit.
- The MOSFETs invert the signals, creating complementary (real and inverted) SPWM signals.

Control and Sequencing with 4017 IC Decade Counter

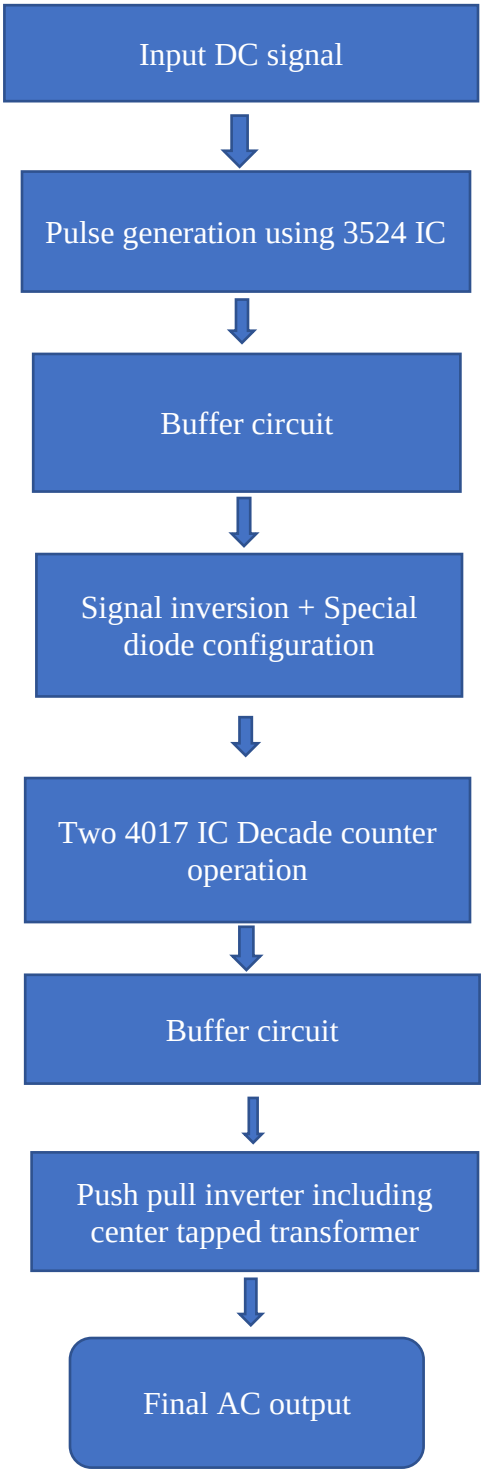
- The real and inverted SPWM signals are provided to the 4017 IC decade counter.
- The decade counter ensures that only one output is active at any moment, effectively sequencing the signals to the push-pull inverter circuit.
- Specific pins (1, 3, 4, 5, and 7) of the 4017 IC are used to generate the SPWM signals, contributing to precise pulse-width modulation.

Push-Pull Inverter Output

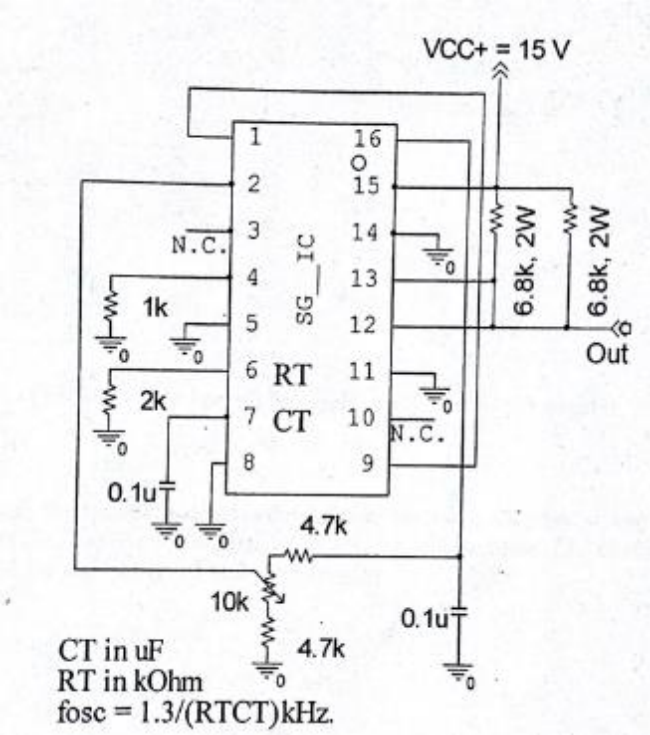
- The sequenced SPWM signals are applied to a push-pull inverter circuit.
- The push-pull configuration ensures efficient power conversion by alternating the flow of current through the transformer, producing an AC output.
- The resulting waveform is a sinusoidal AC signal with reduced harmonic distortion, suitable for powering AC loads.

3.2 Design Method (PO(a))

Block diagram:

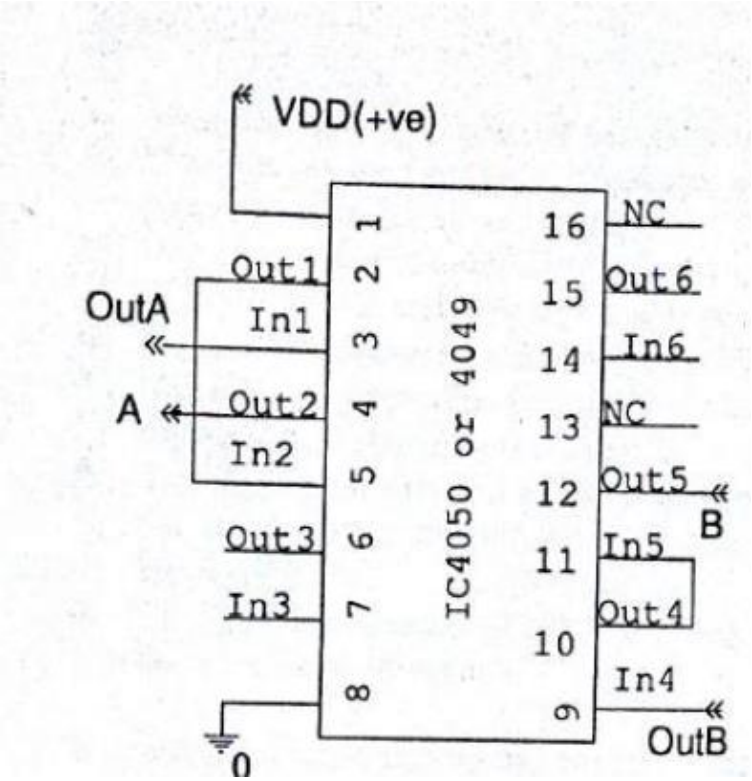


Clock generation:



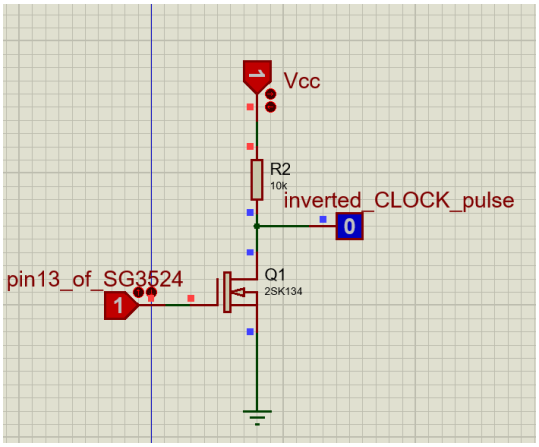
We used this circuit diagram for designing of pulse generation. This is 3524 IC

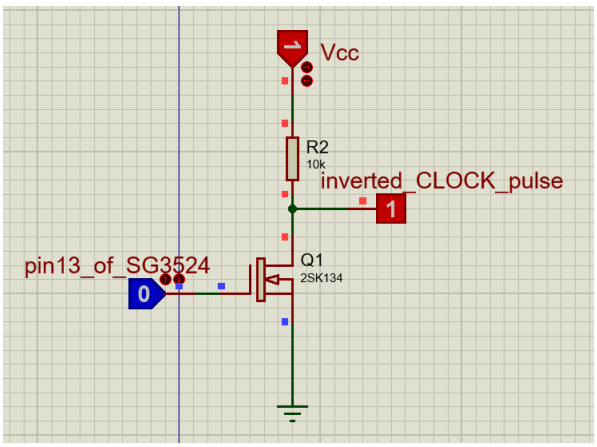
Buffer circuit:



Here we used IC 4050 for the buffer circuit for our project

Clock pulse inversion:



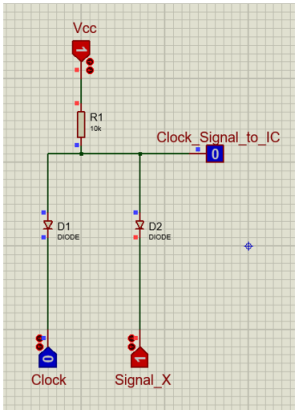


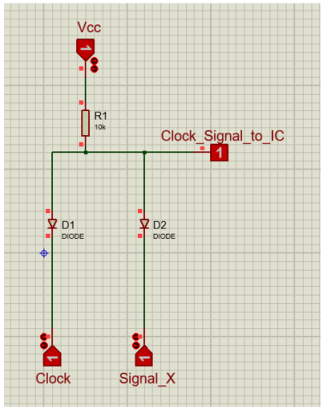
1. **Input Clock Pulse (Pin 13 of SG3524)**:
   The clock pulse from pin 13 of the SG3524 IC is fed to the gate of the MOSFET (Q1, a 2SK134). This signal acts as the control input for the MOSFET.
2. **Gate Control of MOSFET**:
   - When the clock pulse at pin 13 is **low (0)**, the MOSFET remains **off** (non-conducting). As a result, the pull-up resistor (R2, 10k) pulls the output node (inverted_CLOCK_pulse) to **Vcc**, resulting in a **high (1)** output.
   - When the clock pulse at pin 13 is **high (1)**, the MOSFET turns **on** (conducting). This creates a low-resistance path to ground, pulling the output node (inverted_CLOCK_pulse) to **0V** (low output).
3. **Output Inversion**:
   The result is that the output at the node labeled "inverted_CLOCK_pulse" is the logical inversion of the input clock pulse from pin 13 of SG3524.

Summary:

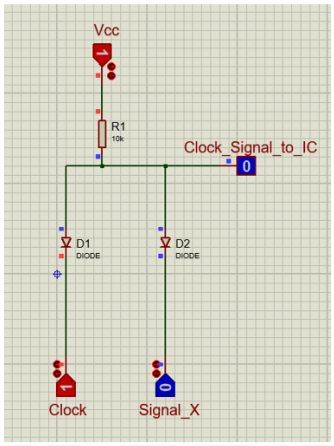
- **Input HIGH**: MOSFET conducts → Output LOW (0V).
- **Input LOW**: MOSFET is off → Output HIGH (Vcc).

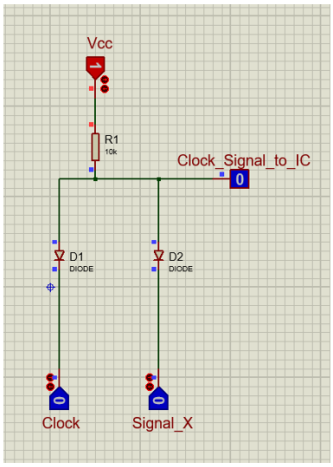
Special diode configuration:





If ‘Signal_X’ is ‘1’ then ‘Clock_signal_to_IC’ will follow the ‘Clock’ signal





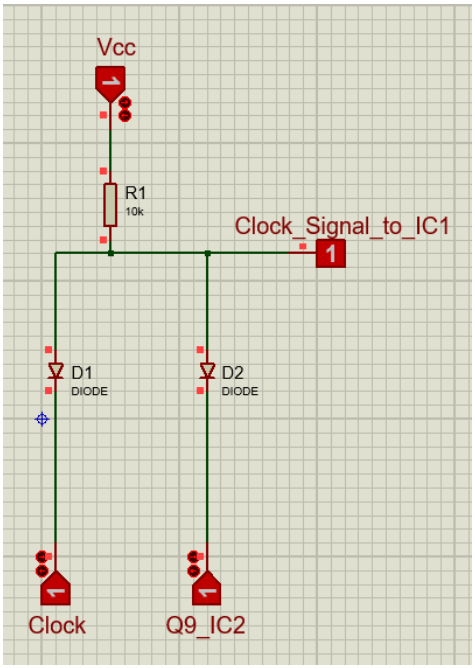
But, if 'Signal_X' is '0' then 'Clock_signal_to_IC' become '0' for any value of 'Clock' signal.
So, we can say,
Clock_signal_to_IC = Clock & Signal_X

Here, we have used **1N4148 diode** as it is capable of fast switching.

Clock Pulse for two 4017 IC1:

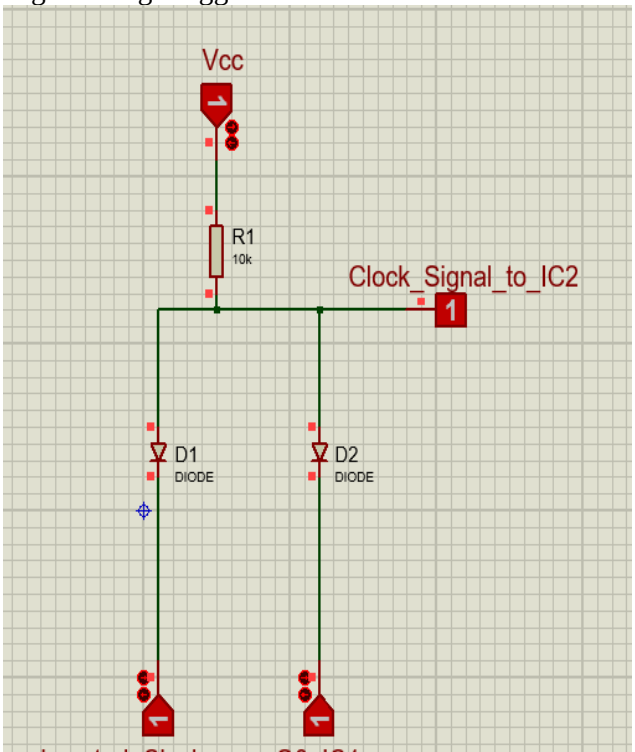
To generate the clock pulse for IC1, we have taken 'Q9' of IC2 as the 'Signal_X'.

So, when 'Q9' of IC2 is '1', IC1 will run. Otherwise, it will hold its previous state.

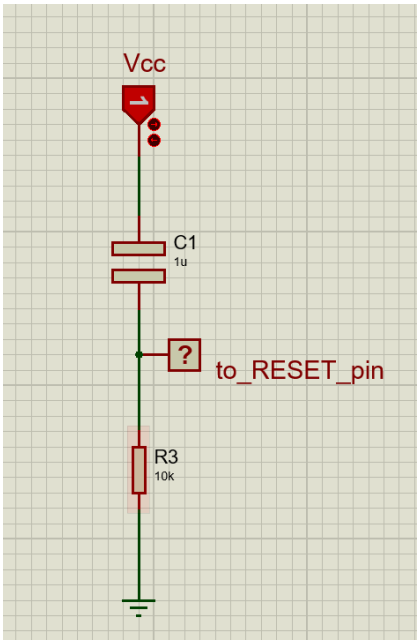


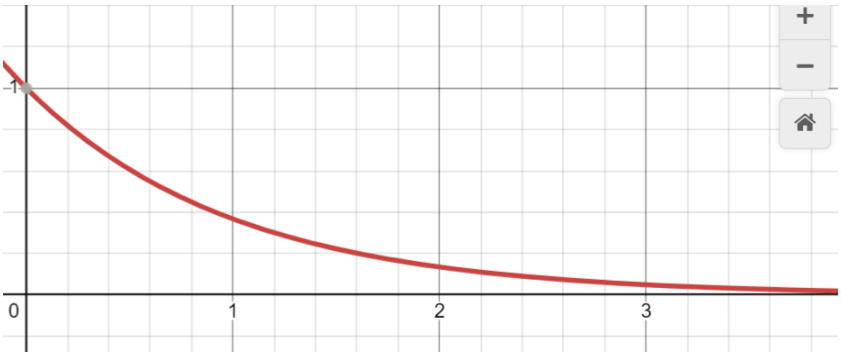
Clock Pulse for 4017 IC2:

To generate the clock pulse for IC2, we have taken 'Q0' of IC1 as the 'Signal_X'.
So, when 'Q0' of IC1 is '1', IC1 will run. Otherwise, it will hold its previous state.
We also use the inverted clock pulse for IC2 to make it negative edge triggered



Capacitor configuration:

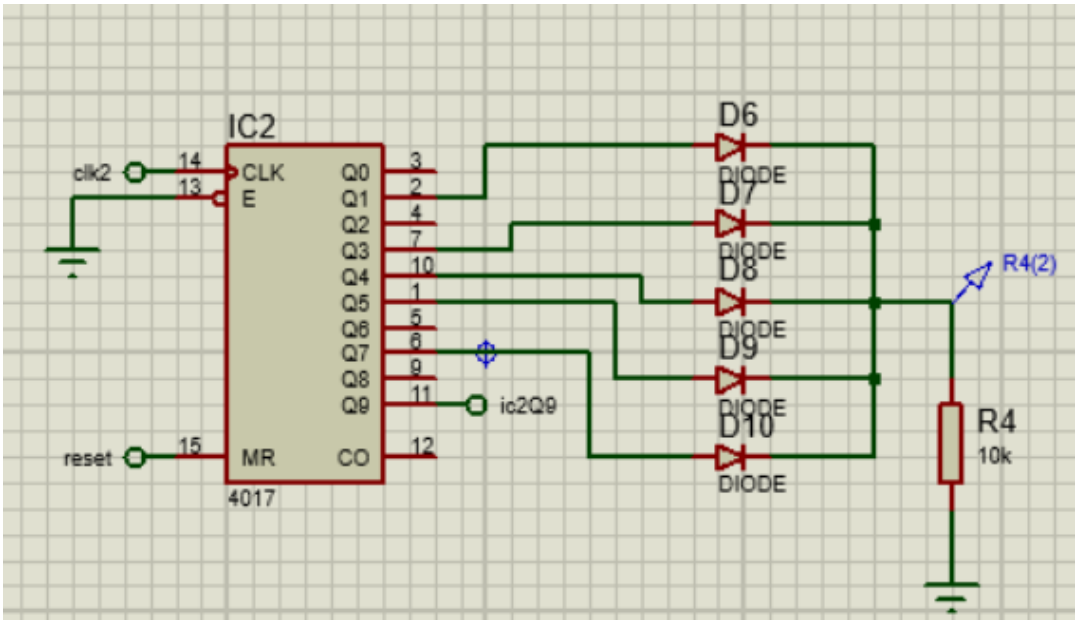




So, after energizing the circuit both IC will be reset first, then both of the IC's Reset pin will be grounded simultaneously.

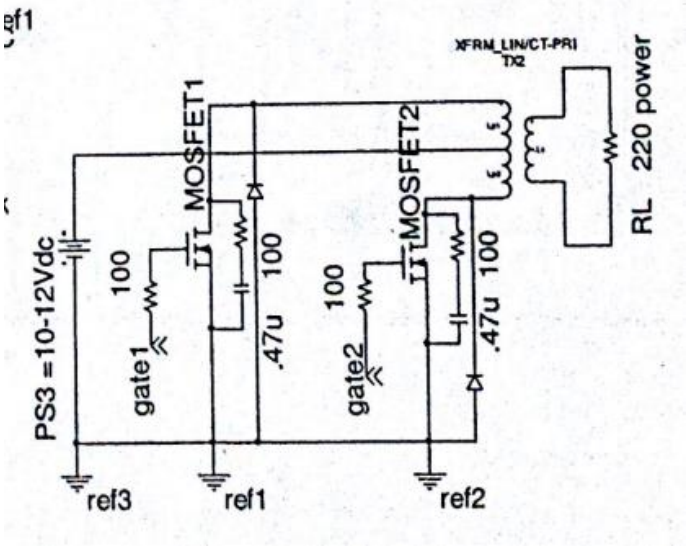
Forming the PWM signal:

From Both ICs we take output 1 345 7 to form the PWM signal

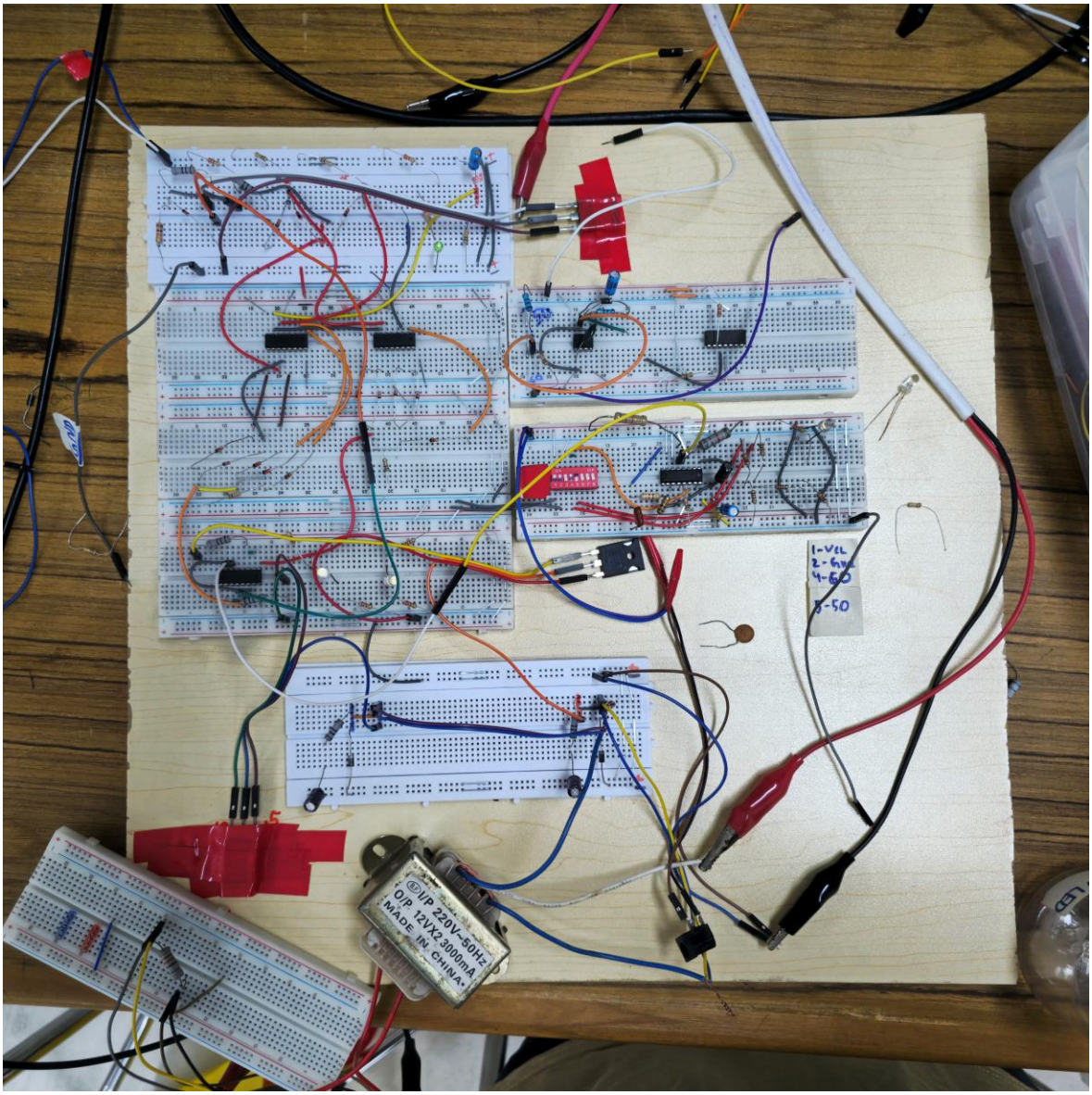


Push-Pull Inverter:

After that, we have used the PWM signals as two ‘gate’ drive signals to drive two MOSFETs of a push-pull inverter.

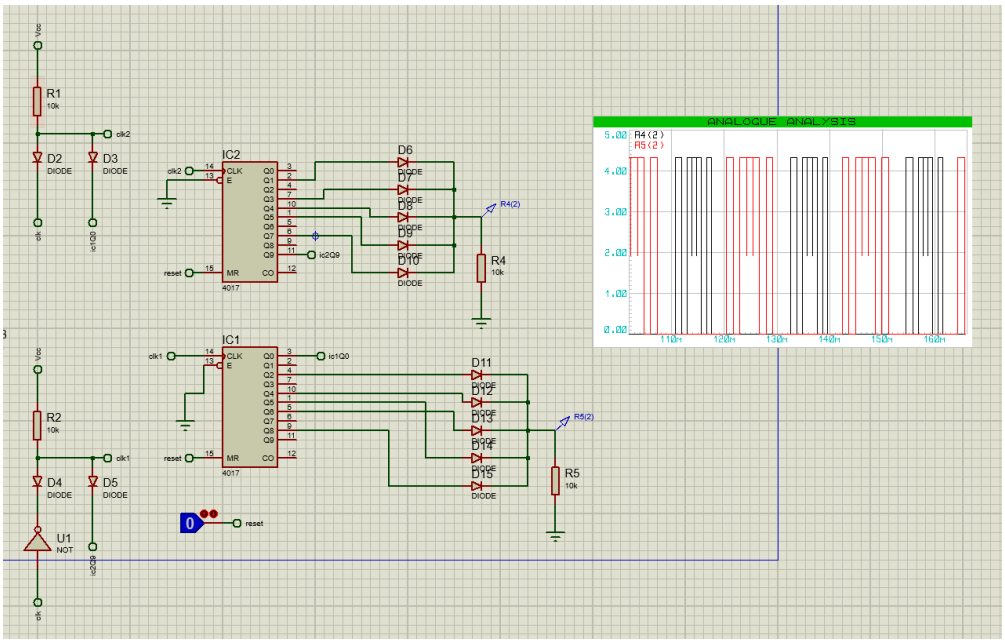


3.3 Circuit Diagram



3.4 Simulation Model

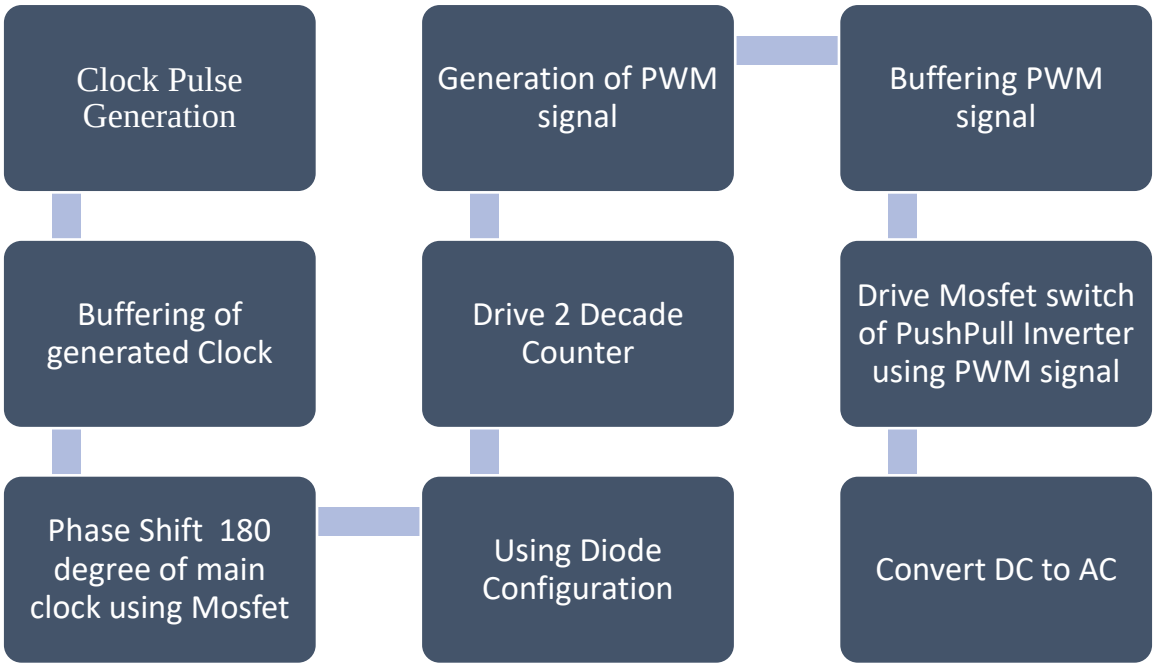
Simulation is conducted on Proteus software:



Special diode configuration and decade counter circuit is simulated here

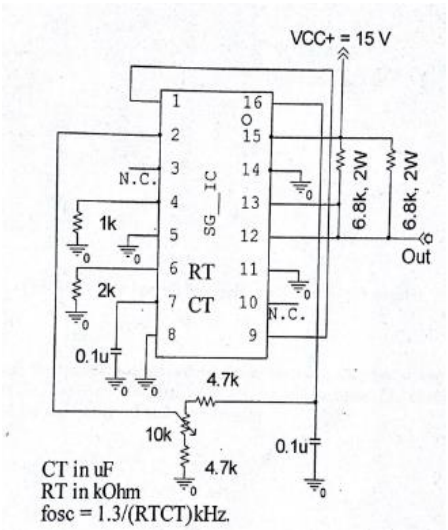
4.Implementation:

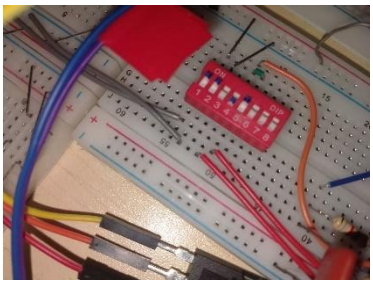
Block diagram of Complete Circuit:



Description:

Clock Generator Circuit:





The **SG3524** is a versatile PWM controller IC widely used in power electronics for regulating and controlling switching devices in applications like inverters, DC-DC converters, and motor drives. It features an internal oscillator, error amplifier, and complementary outputs, simplifying the design of circuits such as push-pull or half-bridge converter.

PIN 6 and 7 is the controlling PIN to control frequency.

For generating main PWM signal of 50 HZ, we have to generate the clock of 1KHz.

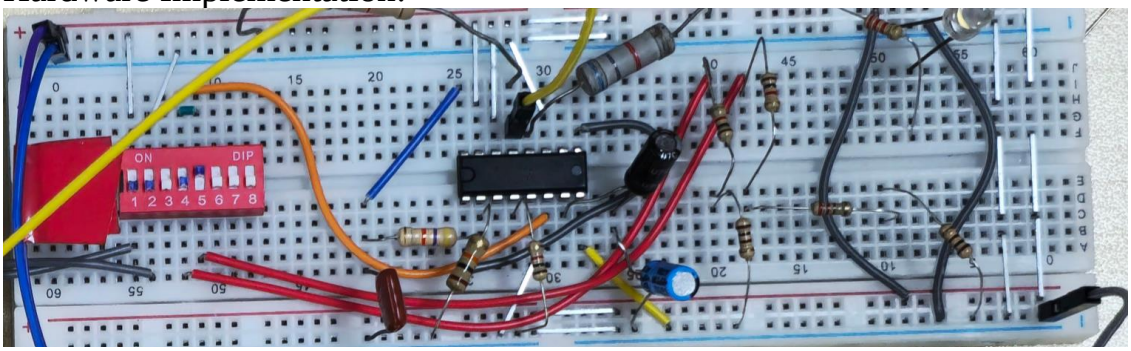
R_T =1.3 Kohm.

Using trial error method C_T =0.32 uF.

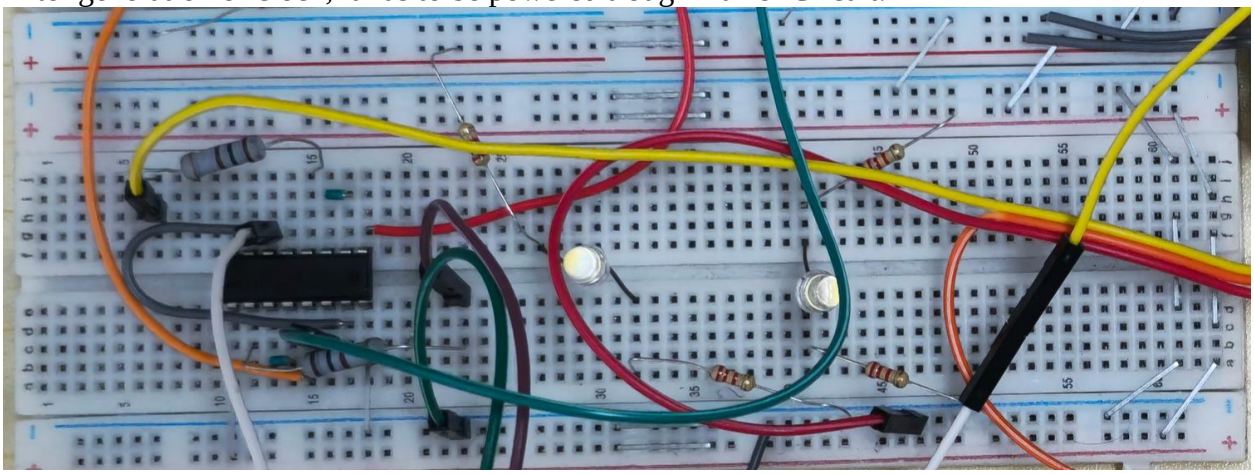
For PWM signal of 60 Hz, Clock frequency=1.2 KHz and R_T =1.083 Kohm

Using DIP switch we can select either 1.3 /1.083 Kohm.

Hardware Implementation:

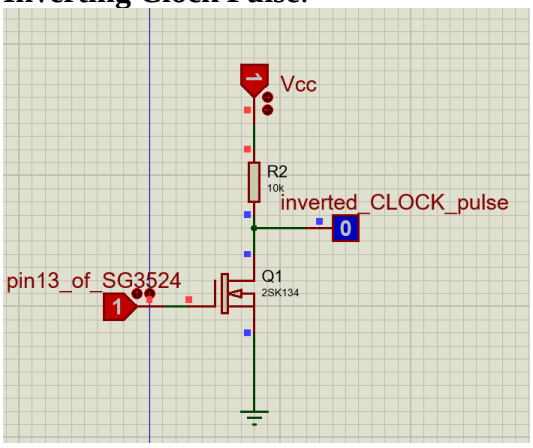


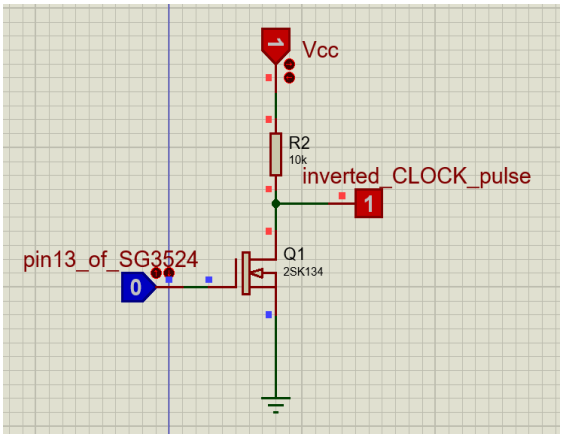
After generation of clock, it has to be powered trough Buffer Circuit.



After Buffering we used a MOSFET for getting 180 degree phase shifted signal.

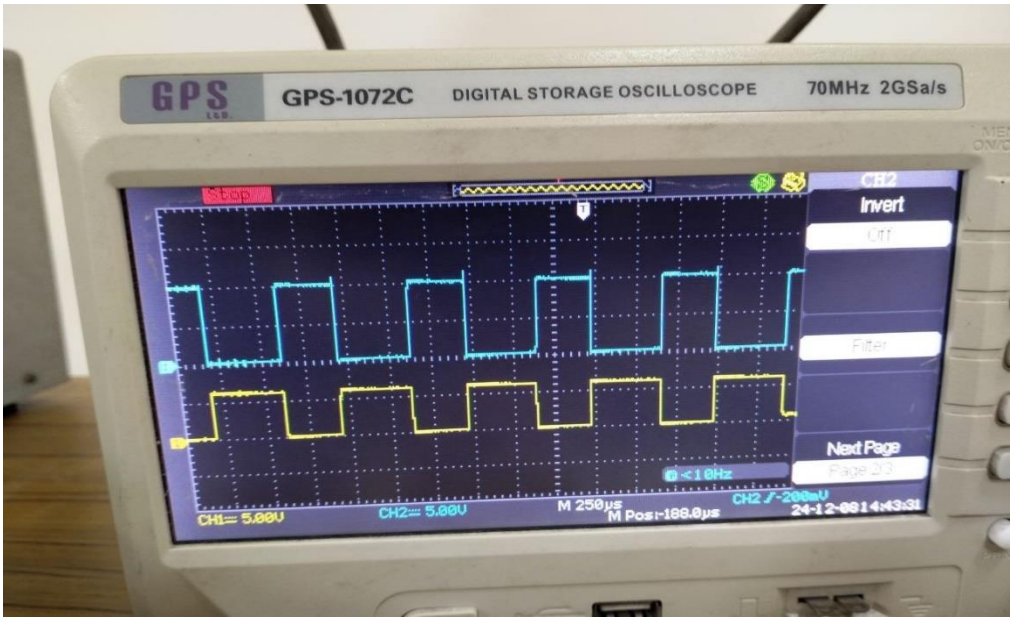
Inverting Clock Pulse:



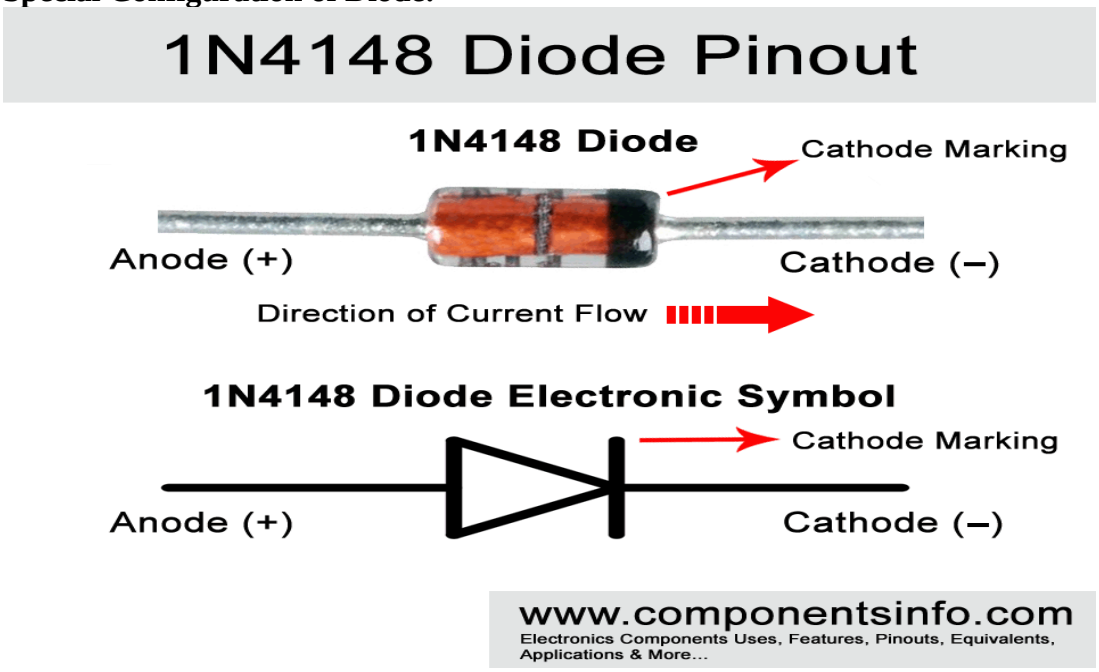


If Pulse is 1, MOSFET switch in On, due to current passing, R2 will dissipate all voltage and we will get 0 voltage Drain side. If, Pulse is zero, Switch is OFF, no current passes and drain side will be Vcc.
Thus Clock is inverted.

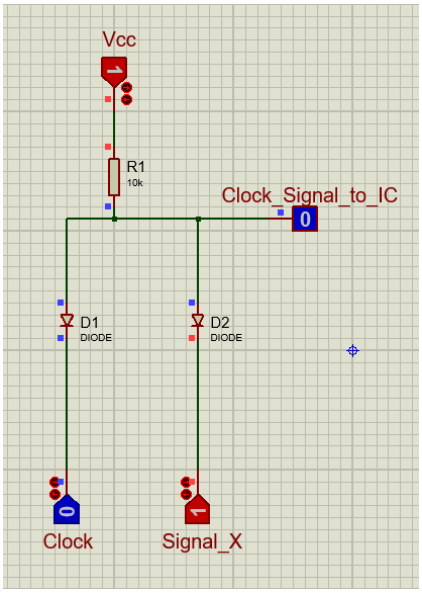
Oscilloscope Output:

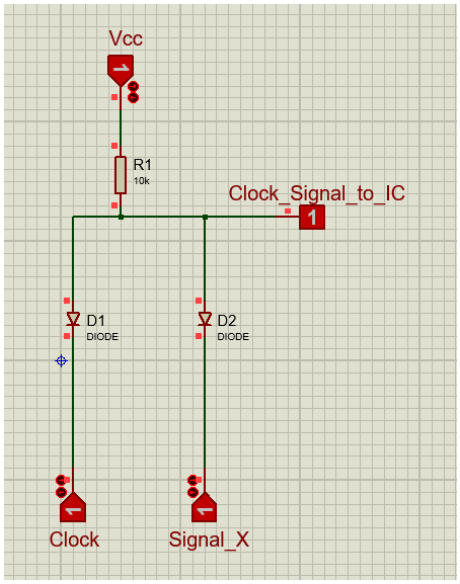


Special Configuration of Diode:



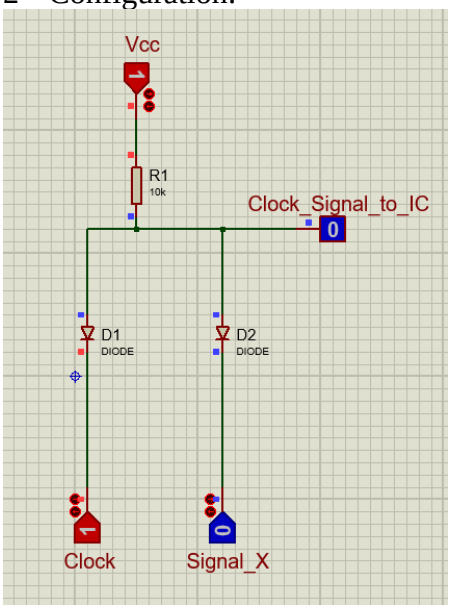
we have used 1N4148 diode as it is capable of fast switching for whole project.

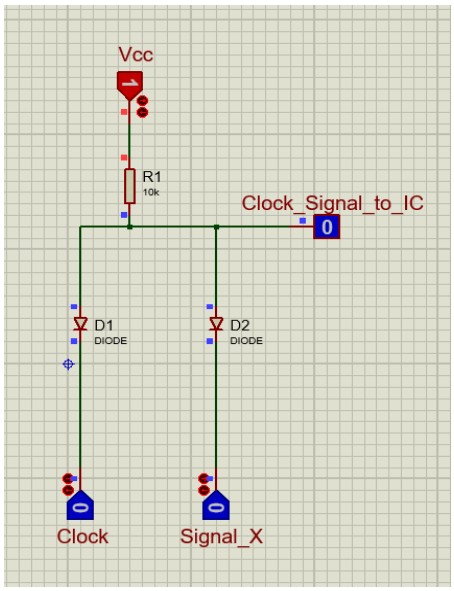




If 'Signal_X' is '1' then 'Clock_signal_to_IC' will follow the 'Clock' signal.
If 'Signal_X' is '1'
And if clock is 0 current passes through Diode 1. Thus, 'Clock_signal_to_IC'=0.
And if clock is 1 current can't pass through D1 or D2. Thus, 'Clock_signal_to_IC'=1.

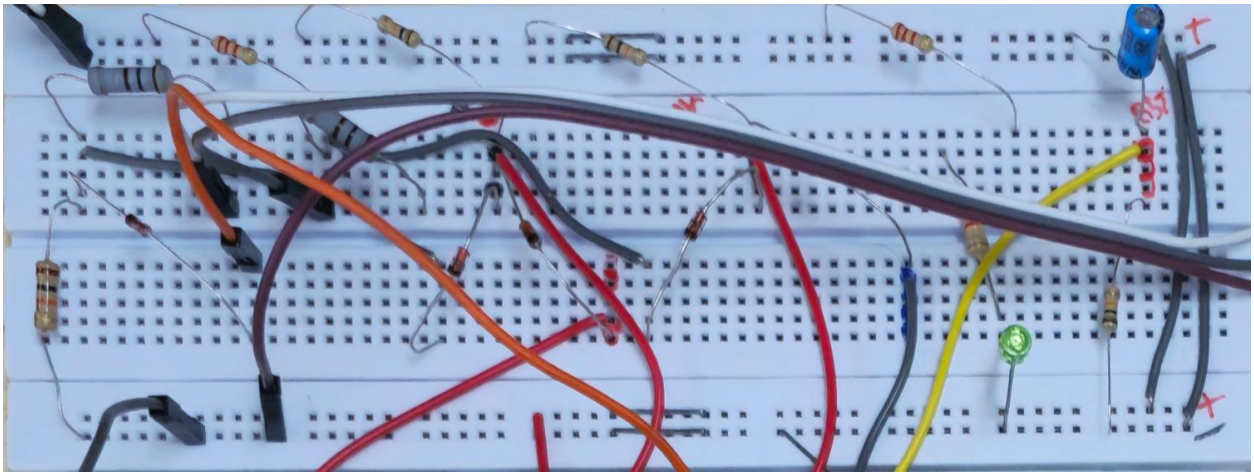
2nd Configuration:





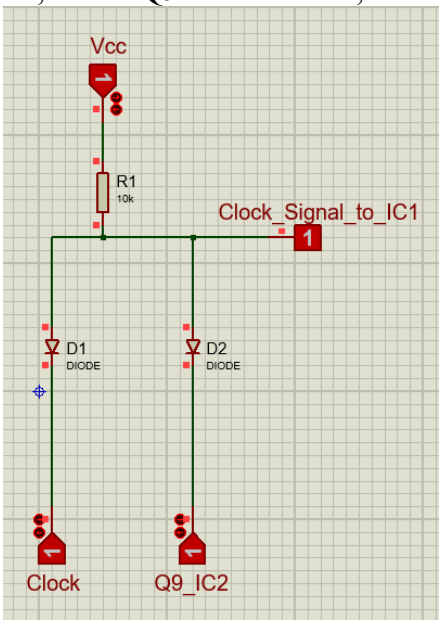
But, if 'Signal_X' is '0' then 'Clock_signal_to_IC' become '0' for any value of 'Clock' signal as current passes through D2.

So, we can say, Clock_signal_to_IC = Clock & Signal_X .
Hardware Setup:



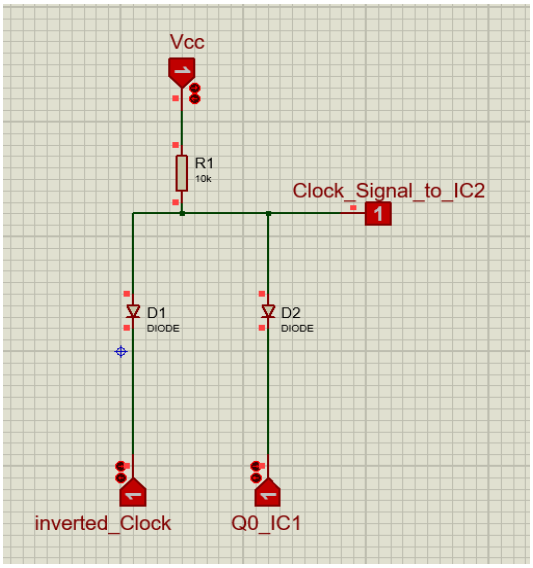
Clock Pulse for two 4017 IC1:

To generate the clock pulse for IC1, we have taken ‘Q9’ of IC2 as the ‘Signal_X’.
So, when ‘Q9’ of IC2 is ‘1’, IC1 will run. Otherwise, it will hold its previous state.



Clock Pulse for 4017 IC2:

To generate the clock pulse for IC2, we have taken ‘Q0’ of IC1 as the ‘Signal_X’.
So, when ‘Q0’ of IC1 is ‘1’, IC1 will run. Otherwise, it will hold its previous state.
We also use the inverted clock pulse for IC2 to make it negative edge triggered.



Visualizing the Working Method of two ICs:

Cycle	Q0 (IC1)	Q9 (IC2)	Clock	IC1 output	Inverted Clock	IC2 output
0	1 **Active: IC2**	0	0	0	1	0
			1		0	
1			0		1	1
			1		0	
2			0		1	2
			1		0	
3			0		1	3
			1		0	
4			0		1	4
			1		0	

Cycle	Q0 (IC1)	Q9 (IC2)	Clock	IC1 output	Inverted Clock	IC2 output
5	1 **Active: IC2**	0	0	0	1	5
			1		0	
6			0		1	6
			1		0	
7			0		1	7
			1		0	
8			0		1	8
			1		0	
	1	1	0		1	
9	0	1	1	1	0	9

Both ICs are active. But IC1 will change first as it is positive edge triggered.

IC2 becomes inactive

Cycle	Q0 (IC1)	Q9 (IC2)	Clock	IC1 output	Inverted Clock	IC2 output
10	0	1 **Active: IC1**	0	1	1	9
			1	2	0	
11			0		1	
			1	3	0	
12			0		1	
			1	4	0	
13			0		1	
			1	5	0	
14			0		1	
			1	6	0	

Cycle	Q0 (IC1)	Q9 (IC2)	Clock	IC1 output	Inverted Clock	IC2 output
15	0	1 **Active: IC1**	0	6	1	9
			1	7	0	
16			0		1	
			1	8	0	
17			0		1	
			1	9	0	
18			0		1	
			1		0	
19	1	0	0	0	1	0
			1		0	

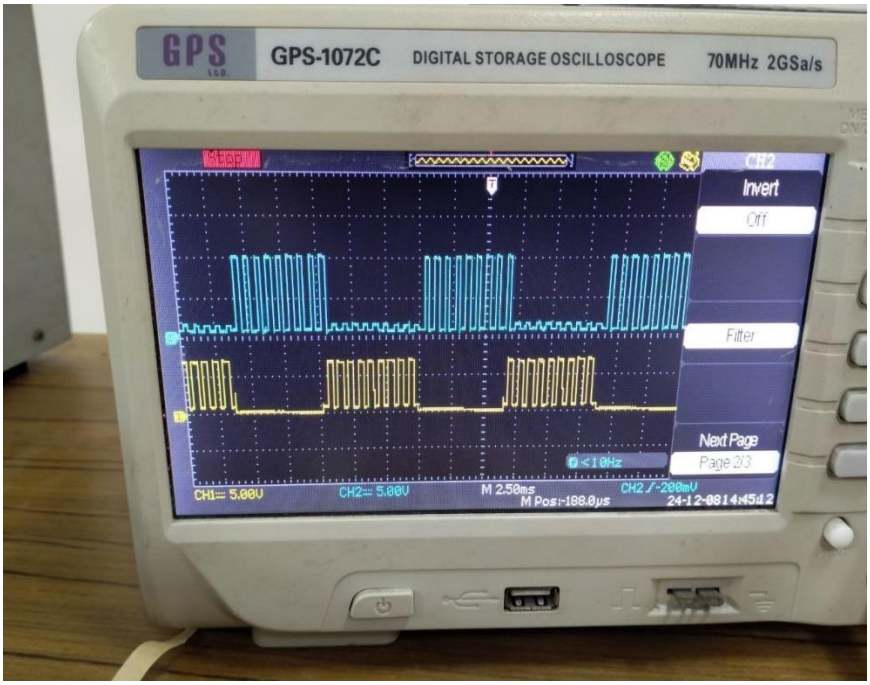
Both ICs are active. But IC2 will change first as it is negative edge triggered.

IC1 becomes inactive

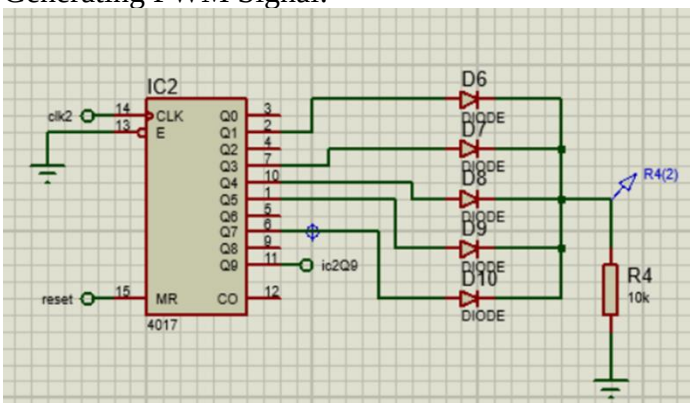
So, we deactivate one IC's clock cycle when other IC's clock is running. Thus, only one IC is running at a time.
Here, 20 clock pulses form
one full cycle of the sinusoidal output.
Therefore, we take 1kHz clock for 50Hz output,
and 1.2kHz for 60Hz output

Clock Pulse for IC 4017: Decade Counter

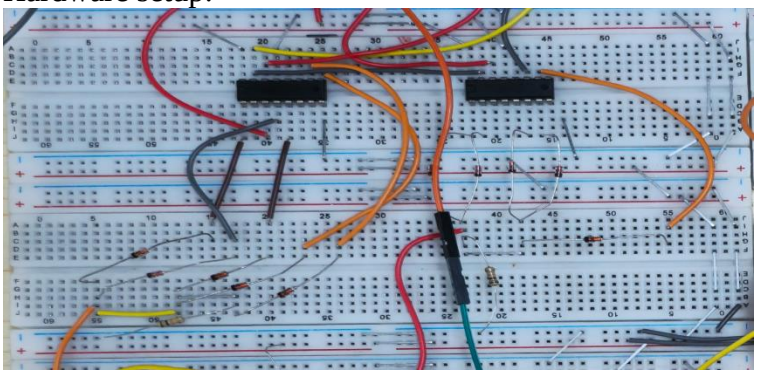
Due to special configuration of Diode, when IC1 will get positive edge, IC2 will be off.
Vice versa for IC2



Generating PWM Signal:



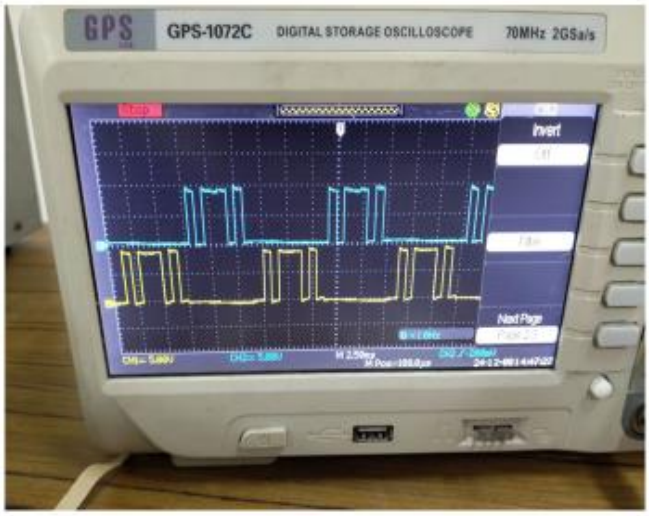
Hardware setup:



After getting positive edge, from Both ICs we take output 1 345 7 to form the PWM signal. PWM signal generally, is an un-uniform width signal. Middle Pulse should be widened than the 2 marginalized pulse with less width.

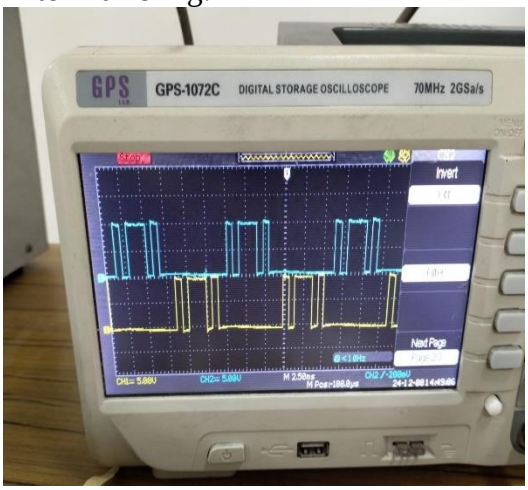
Oscilloscope Output:

PWM Signal (Oscilloscope output):



Then, we have passed the PWM signals through a buffer 4050 IC to prepare it for driving MOSFETs

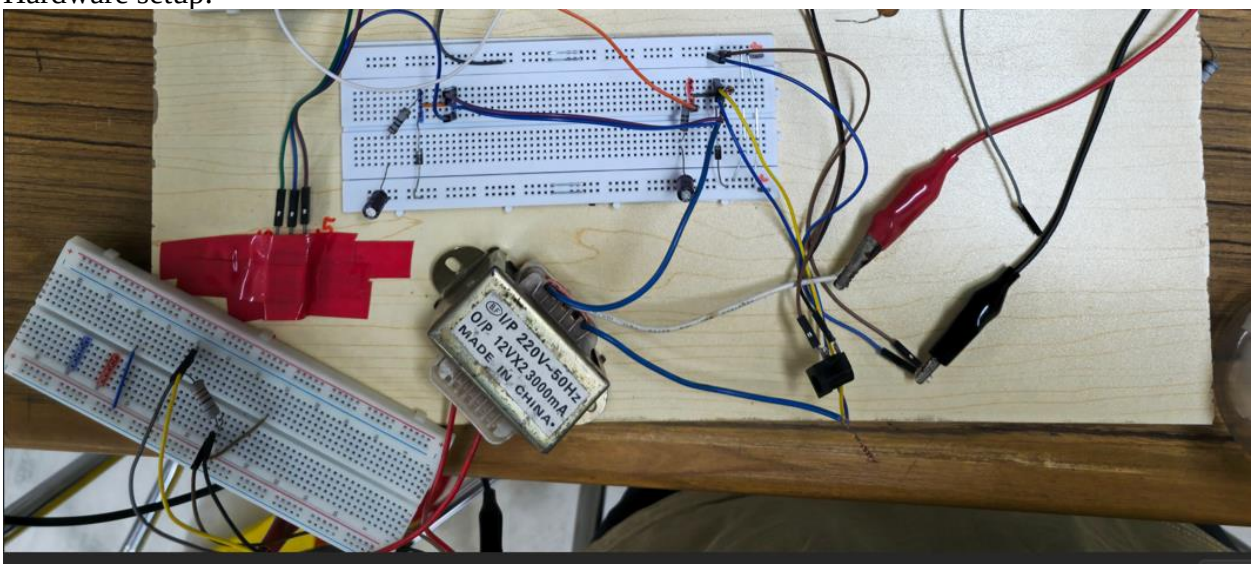
After Buffering:



Push-Pull Inverter:

A **push-pull inverter** is a power electronics circuit topology widely used for converting DC input into AC output with high efficiency and reliability. It operates by using a center-tapped transformer and two active switching devices, such as MOSFETs or BJTs, which alternately drive the transformer primary winding. This alternation creates an alternating magnetic field in the transformer, producing a sinusoidal or square wave AC output at the secondary winding.

Hardware setup:

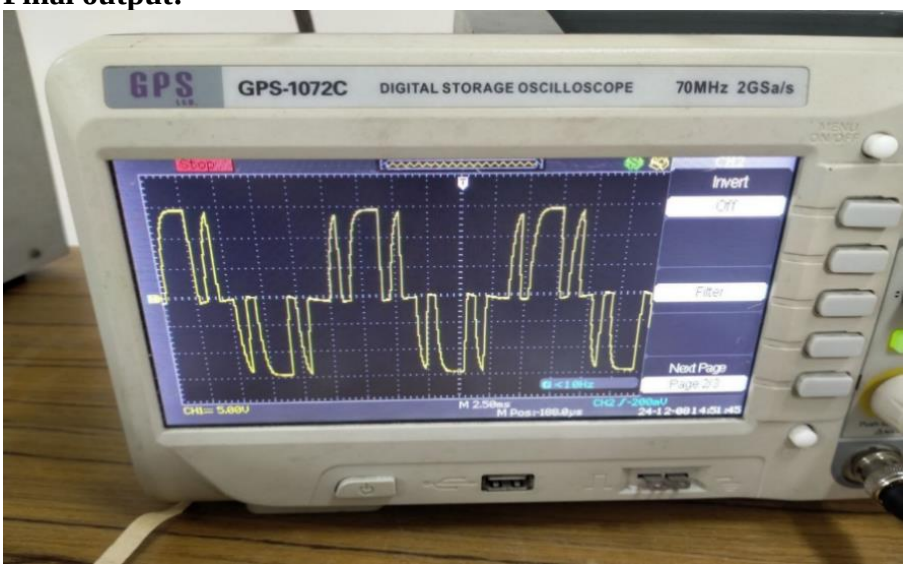


A push-pull inverter works by converting DC voltage into AC through alternating operation of two switches and a center-tapped transformer. The DC source is connected to the center tap of the transformer's primary winding, while the two ends of the winding are connected to switching devices like MOSFETs or BJTs.

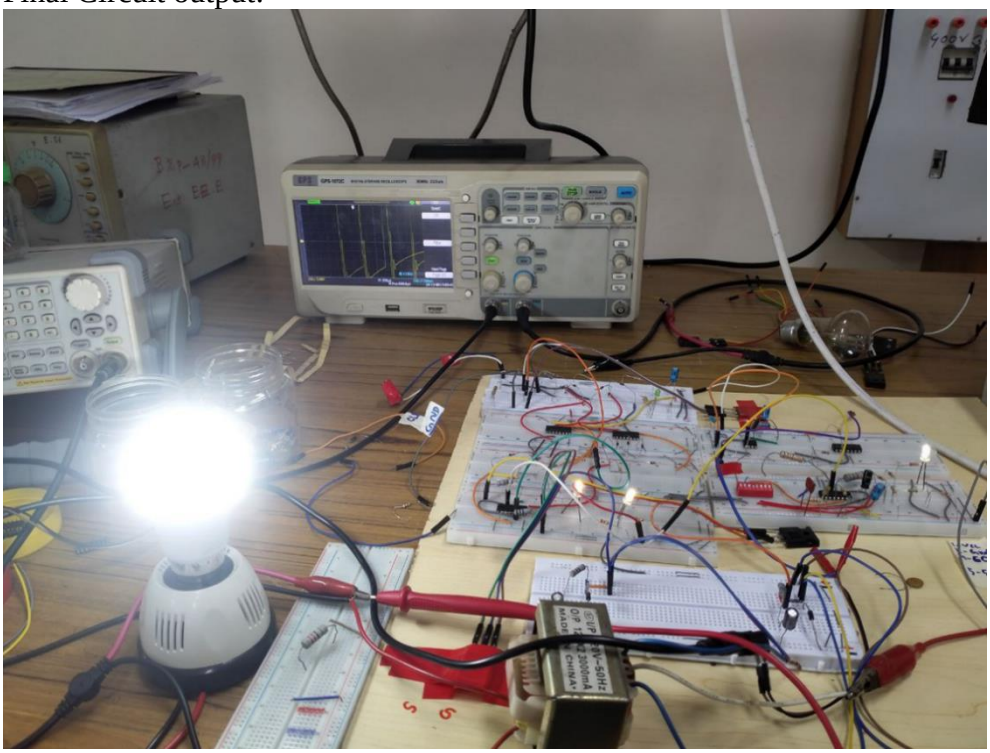
A control circuit alternates the switching of the transistors. When one transistor is turned on, current flows through one half of the primary winding, creating magnetic flux in one direction. When the other transistor turns on, the current flows through the other half of the winding, reversing the magnetic flux. This alternating flux induces an AC voltage in the transformer's secondary winding. The transformer's turn ratio determines the voltage level of the output, and filters can be added for waveform smoothing if a pure sine wave output is needed.

Complete Hardware setup:

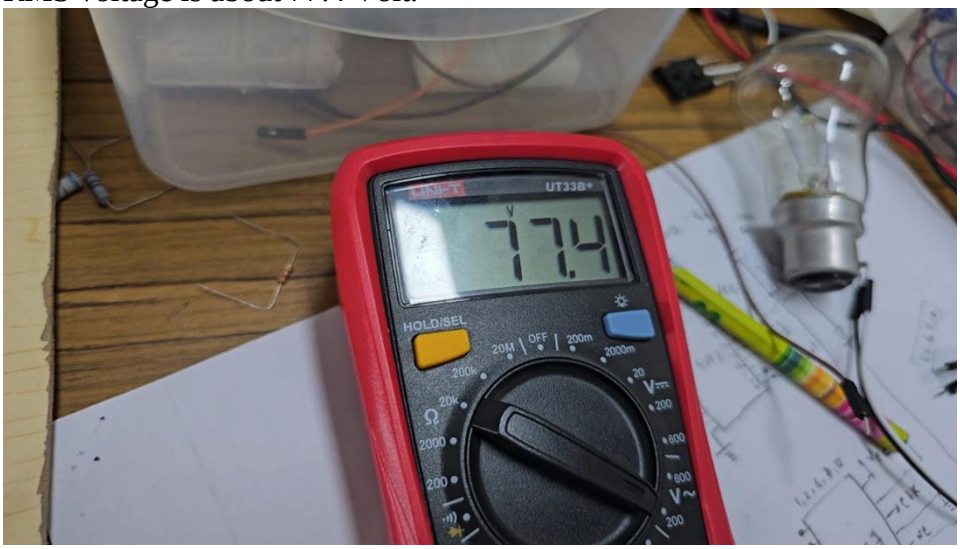
Final output:



Final Circuit output:



RMS voltage measured with Multi-meter:
RMS voltage is about 77.4 Volt.



5 Design Analysis and Evaluation

5.1 Novelty

- The SPWM inverter design demonstrates a unique approach by avoiding the direct generation of sine and triangular waves or the use of comparators, typically found in conventional SPWM designs.
- The innovation lies in the use of decade counters (4017 ICs) to generate un-uniform pulse width signals for MOSFET gates, simplifying the circuitry and reducing complexity.
- A novel diode configuration manages clock signals, ensuring only one decade counter operates at a time, thus improving synchronization and reducing errors.
- The design achieves a sine wave approximation with minimal components and avoids reliance on expensive sine wave generators or complex control mechanisms.

Below is a detailed filling of the specified points based on the SPWM inverter project presentation uploaded:

5.2 Design Considerations (PO(c))

5.2.1 Considerations to Public Health and Safety

- The project ensured the safe operation of the inverter circuit by selecting components like 1N4148 diodes for fast switching and MOSFETs with appropriate ratings to prevent overheating or failure.
- Protective measures, such as proper insulation and grounding, were considered to avoid electrical hazards during circuit operation.

5.2.2 Considerations to Environment

- The inverter design avoided the use of toxic materials and minimized energy losses by optimizing the SPWM waveform generation, thus reducing heat dissipation and conserving energy.
- Use of efficient MOSFETs ensures lower power consumption, contributing to environmental sustainability.

5.2.3 Considerations to Cultural and Societal Needs

- The inverter design supports energy-saving and renewable energy systems like solar power setups, addressing the increasing societal demand for clean energy solutions.

- The cost-effective and efficient inverter can provide electricity to remote and underprivileged areas where grid access is limited.

5.3 Investigations (PO(d))

5.3.1 Design of Experiment

- The experiment involved generating clock pulses, configuring decade counters (4017 ICs), and designing a push-pull inverter for SPWM signal application to MOSFETs.

5.3.2 Data Collection

- Oscilloscope readings were taken to analyze the clock pulses and the PWM signals before and after buffering.
- Voltage and frequency measurements were recorded to ensure the inverter output closely resembled a sine wave.

5.3.3 Results and Analysis

- Observed that the PWM signals formed using the 4017 ICs successfully drove the MOSFETs.
- The final output approximated a sine wave with an RMS value of 77.4 V, meeting the project's design goal.

5.3.4 Interpretation and Conclusions on Data

- The results validated that the SPWM technique, implemented through decade counters, can achieve sinusoidal outputs without direct generation of sine or triangular waves.

5.4 Limitations of Tools (O(e))

- **Oscilloscope:** Limited bandwidth may affect the accuracy of high-frequency signal visualization.
- **Decade Counters (4017 ICs):** Prone to errors if the clock signal is not stable or correctly inverted.
- **MOSFETs:** May overheat if the gate drive signal is not properly buffered.
- **Tolerance Issues:** Components like resistors and capacitors have tolerance ranges that can slightly affect frequency accuracy.

5.5 Impact Assessment (PO(f))

5.5.1 Assessment of Societal and Cultural Issues

- The inverter promotes access to reliable and affordable power, enhancing societal well-being, especially in rural or low-income communities.

5.5.2 Assessment of Health and Safety Issues

- Proper insulation and adherence to electrical safety standards ensured the safety of users and operators.

5.5.3 Assessment of Legal Issues

- The project complied with intellectual property regulations by designing a novel solution and avoiding patented techniques.

5.6 Sustainability Evaluation (PO(g))

- The design supports the integration of renewable energy sources like solar panels.
- Energy-efficient operation reduces waste heat, promoting long-term environmental sustainability.
- The use of commonly available components ensures economic feasibility and resource sustainability.

5.7 Ethical Issues (PO(h))

- Ethical principles were upheld by ensuring that the inverter design was safe, cost-effective, and environmentally friendly.
- Proper credit was given to sources of theoretical knowledge used in the design.
- Ethical challenges, such as ensuring the accuracy of results, were mitigated by thorough testing and validation of the circuit under various conditions.

6. Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member:

ID	work
2006099	Literature review , Designing, Simulation, Hardware assembling, Circuit test, Report writing
2006117	Literature review, component ordering and buying, hardware assembling, Circuit test, Report writing
2006118	component ordering and buying ,Hardware assembling, Circuit test, Report writing
2006121	Designing, Simulation, Hardware assembling, Circuit test.

6.2 Mode of Team-Work

Final assemble
Report writing
Finding the proper solution when we change our procedure

6.3 Diversity Statement of Team

Our team recognizes that diversity in both skills and perspectives is essential to fostering innovation and achieving success in any project. We believe that a team comprised of individuals from different backgrounds, experiences, and areas of expertise enriches the problem-solving process and enhances creativity. By bringing together people with varying perspectives, we can approach challenges from multiple angles, ensuring that our design process is thorough, inclusive, and adaptable.
The diversity within our team includes various technical skill sets, ranging from engineering knowledge and Hardware and simulation skills to system design and analysis. Additionally,

our team members come from different academic and cultural backgrounds, allowing us to appreciate diverse viewpoints and approaches in tackling the project. This diversity not only enhances the overall quality of our work but also aligns with our goal to create solutions that can be understood, applied, and appreciated by a wide audience.
We are committed to promoting an inclusive environment where all team members feel valued and respected. By embracing our differences, we can foster collaboration and work together to create impactful, innovative solutions that meet the needs of our project while considering the broader societal and cultural context

6.4 Log Book of Project Implementation

Date	Milestone achieved
8/11/24	Simulation of various Circuit.
12/11/24	Built a rough circuit
15/11/24	1st Circuit test of various part
18/11/24	Complete Circuit built.
19/11/24	1st complete Circuit test and found bugs.
20/11/24	Tried to Debug
04/12/24	Shifted to SG3524 and buffer for better pulse
05/12/24	Got PWM Signal
07/12/24	Got final output using push-pull inverter.
08/12/24	Final project was done

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

This project presents the design and implementation of a Sinusoidal Pulse Width Modulation (SPWM) DC-to-AC inverter using a 4017 IC decade counter. A pure DC input is modulated using the SG3524 IC to generate PWM signals, which are smoothed by a buffer circuit and inverted using MOSFETs. The inverted and non-inverted signals are fed into the 4017 IC, ensuring only one active output at a time for precise control. The SPWM signals are then applied to a push-pull inverter circuit, generating an AC output with minimal harmonic distortion. This design achieves efficient, cost-effective DC-to-AC conversion, making it suitable for

applications such as renewable energy systems and uninterruptible power supplies while offering a robust educational model.

7.2 User Manual

Introduction

This manual provides guidance on operating the SPWM DC-to-AC inverter designed using the SG3524 and 4017 ICs. The system converts a DC input into an AC output using SPWM for high-quality power conversion.

Setup Instructions

1. **Power Supply**: Connect a pure DC source to the input terminals. Ensure the voltage matches the design specifications.
2. **Connections**: Verify all connections, including the SG3524 IC, buffer circuit, MOSFETs, and 4017 IC. Double-check the push-pull inverter connections.
3. **Load Connection**: Attach the desired AC load to the output terminals.

Operation

1. Power ON the DC input supply.
2. The SG3524 IC will generate PWM signals, which are processed and inverted by the circuit.
3. The 4017 IC controls the sequencing, and the push-pull inverter delivers an AC output to the load.

Safety Precautions

- Avoid exceeding the rated DC input voltage.
- Ensure proper cooling for MOSFETs to prevent overheating.
- Disconnect power before making modifications.

Maintenance

Periodically check for loose connections and component integrity.

7.3 Github Link

The onedrive link of the simulation model is given here:

EEE-316_PROJECT_GROUP-6

“ https://buetedu-my.sharepoint.com/personal/2006117_eee_buet_ac_bd/_layouts/15/onedrive.aspx?id=%2Fpersonal%2F2006117%5Feee%5Fbuet%5Fac%5Fbd%2FDocuments%2FEEE%2D316%5FPROJECT%5FGROUP%2D6&ga=1 “

7.4 YouTube Link

The onedrive link for video demonstration is given here:

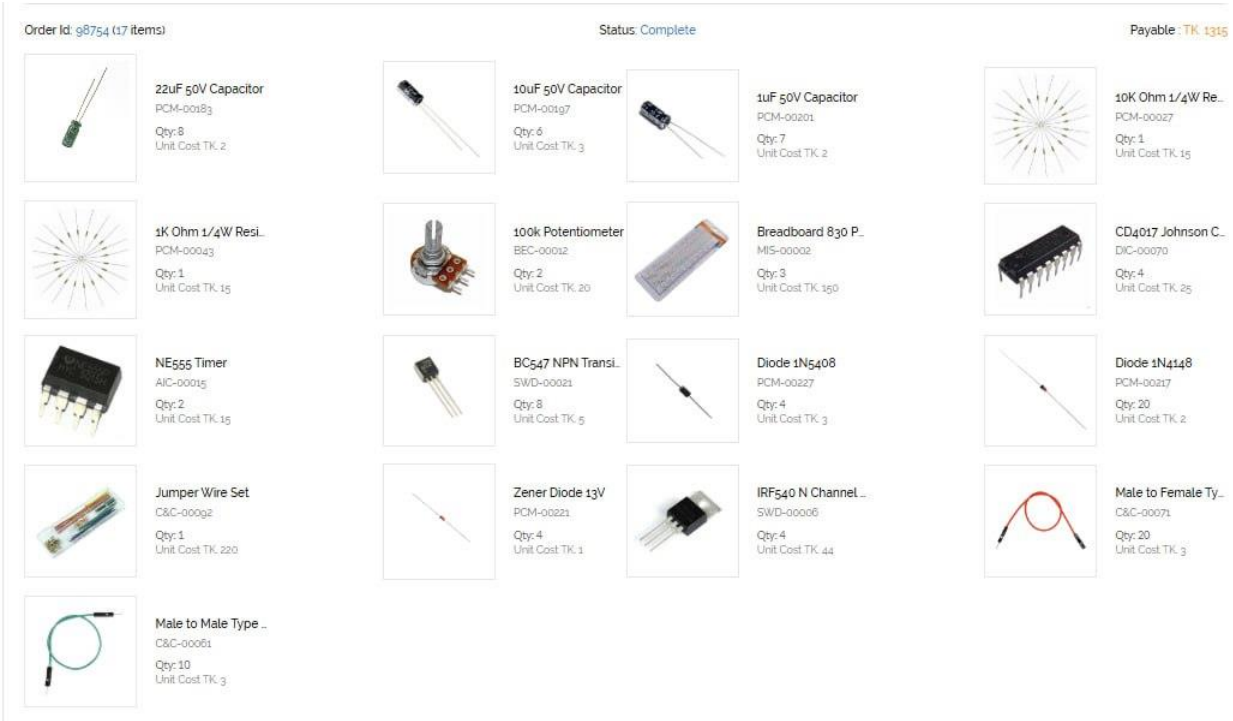
EEE-316_PROJECT_GROUP-6

“ https://buetedu-my.sharepoint.com/personal/2006117_eee_buet_ac_bd/_layouts/15/onedrive.aspx?id=%2Fpersonal%2F2006117%5Feee%5Fbuet%5Fac%5Fbd%2FDocuments%2FEEE%2D316%5FPROJECT%5FGROUP%2D6&ga=1 “

8 Project Management and Cost Analysis (PO(k))

8.1 Bills of Materials

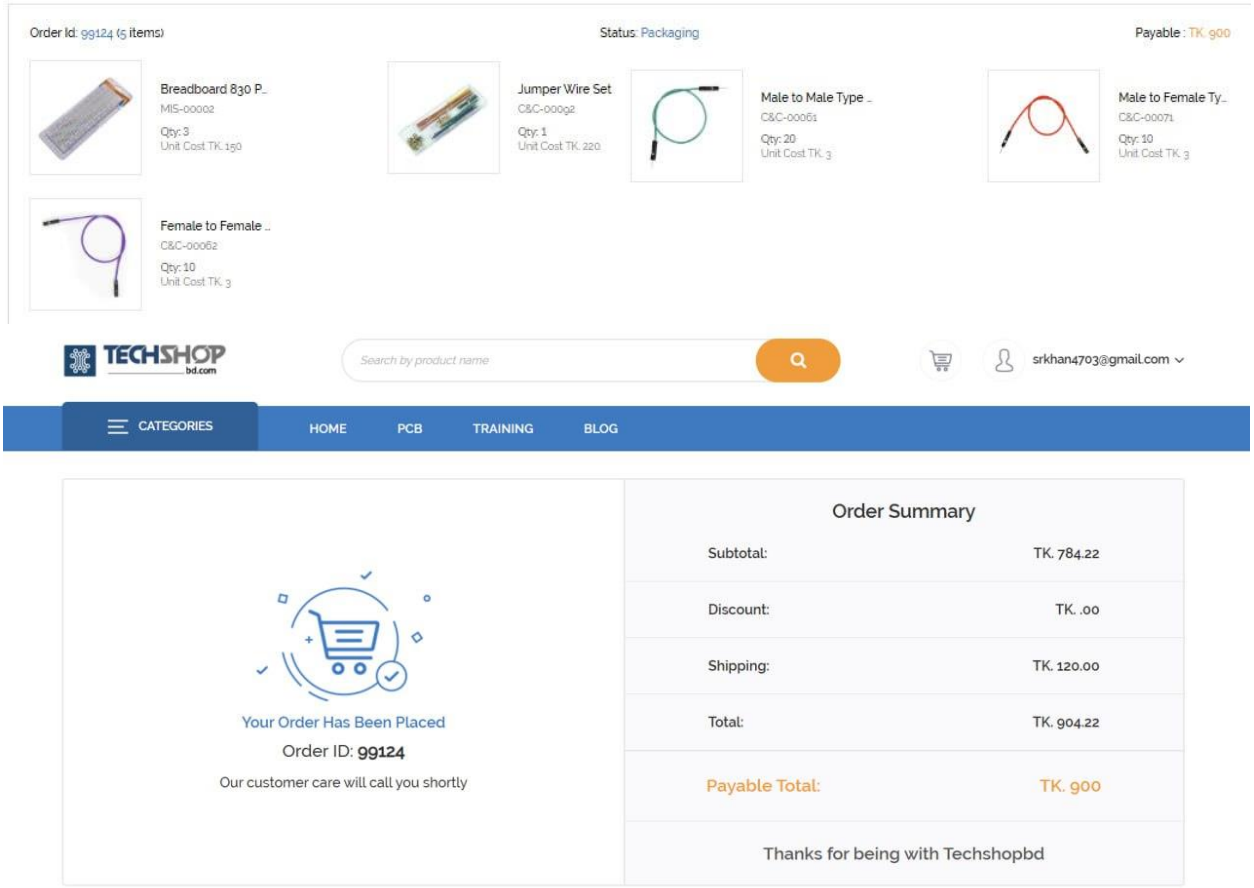
Date:08/11/24



Date:26/11/24

Image	Product Name	Model	Quantity	Unit Price	Total
	74HC08 / 74LS08 / 7408 QUAD AND GATE, 2I/P, DIP-14 And Gate	0659	2	৳32.00	৳64.00
	7474 / 74HC74 / 74LS74 DUAL D-TYPE FLIP-FLOP DIP14	0654	4	৳33.00	৳132.00
	74HC32 / 74LS32 / 7432 QUAD OR GATE, 2I/P, DIP-14	0643	1	৳22.00	৳22.00
				Sub-Total	৳218.00
				Total	৳218.00

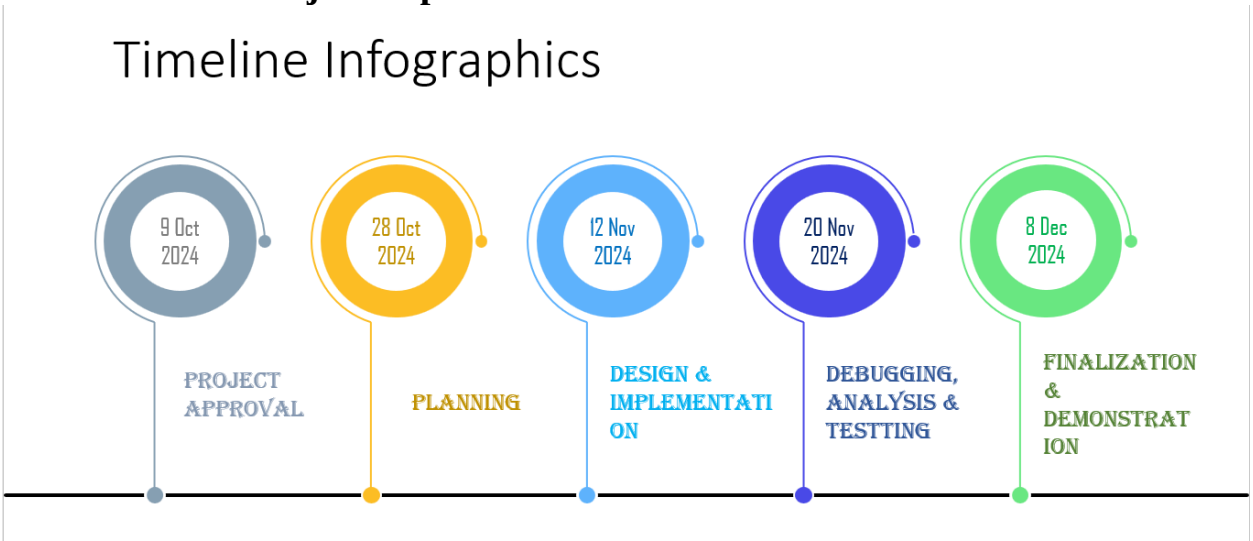
Date:03/12/24



8.2 Cost analysis

Component	Price
Bread board	600
connector wire	650
Resistor and pot	70
capacitor	50
ICs	320
transformer	250
Doide	65
Mosfet	180
Some extra cost	200
Shipping cost	300

8.3 Timeline of Project Implementation



9 Future Work (PO(l))

1. **Improving Output Waveform Quality:**
    - Explore advanced SPWM techniques to generate a waveform that more closely approximates a pure sine wave.
    - Introduce filters to smoothen the output and minimize harmonic distortion.
2. **Integration with Renewable Energy Systems:**
    - Adapt the inverter to interface with solar photovoltaic systems, enhancing its applicability in sustainable energy solutions.
3. **Enhancing Efficiency:**
    - Investigate the use of more advanced MOSFETs with lower switching losses to improve the efficiency of the inverter.
    - Optimize the clock pulse and PWM generation circuits to further reduce energy losses.
4. **Design Scalability:**
    - Scale the design for higher power applications, ensuring robustness under heavy loads by using advanced power electronics components.
5. **Automated Control:**
    - Integrate a microcontroller or FPGA to dynamically control the inverter's frequency and voltage output for different use cases.
6. **Protection Features:**
    - Incorporate over-voltage, over-current, and thermal protection mechanisms to enhance the safety and durability of the inverter.
7. **Miniaturization:**
    - Develop a compact version of the circuit for portable and embedded applications.

10 References

1. **Lecture Notes:** Course materials from EEE 316, Bangladesh University of Engineering and Technology (BUET).
2. **Datasheets:**
    - 4017 Decade Counter IC Datasheet.
    - 1N4148 Fast Switching Diode Datasheet.
    - MOSFET and Buffer IC 4050 Datasheets.
3. **Books and Articles:**
    - B.K. Bose, *Power Electronics and Motor Drives.*
    - Muhammad H. Rashid, *Power Electronics: Circuits, Devices, and Applications.*
4. **Web Resources:**
    - Tutorials on SPWM inverter design from electronics and power engineering forums.
    - Application notes on PWM signal generation and inverter design from component manufacturers.
5. **Experimental Results:** Oscilloscope readings and other measurements recorded during project implementation.